

Christopher Musco

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Academic Positions

New York University <i>Assistant Professor, Tandon School of Engineering</i> <i>Department of Computer Science and Engineering</i>	New York, NY 2019 - present
Microsoft Research <i>Visiting Researcher, Machine Learning and Optimization Group</i>	Redmond, WA Summer 2020
Princeton University <i>Research Instructor, Department of Computer Science</i>	Princeton, NJ 2018 – 2019
Massachusetts Institute of Technology <i>Research and Teaching Assistant, Theory of Computation Group</i>	Cambridge, MA 2013 – 2018
Yahoo Labs <i>Research Intern, Scalable Machine Learning Group</i>	New York, NY Summer 2015

Research Areas

foundations of data science • scalable machine learning • numerical linear algebra • theory of algorithms
randomization • dimensionality reduction • sketching & streaming • spectral graph algorithms • optimization

Education

Massachusetts Institute of Technology <i>Ph.D. in Computer Science</i> Advisor: Jonathan Kelner Thesis: Faster Linear Algebra for Data Analysis and Machine Learning M.S. Thesis: Dimensionality Reduction for Sparse and Structured Matrices	Cambridge, MA 2013 – 2018
Yale University <i>B.S. in Applied Mathematics, B.S. in Computer Science</i>	New Haven, CT 2008 – 2012

Publications

*Authors appear in alphabetical order, in the tradition of mathematics and theoretical computer science. Any exceptions are marked with a *, which indicates that authors are ordered by contribution.*

Fast and Near-Optimal Diagonal Preconditioning. Arun Jambulapati, Jerry Li, Christopher Musco, Aaron Sidford, Kevin Tian. *Preprint* 2021.

Finding the Mode of a Kernel Density Estimate. Jasper C.H. Lee, Jerry Li, Christopher Musco, Jeff M. Phillips, Wai Ming Tai. *European Symposium on Algorithms (ESA)*. 2021.

Active Sampling for Linear Regression Beyond the ℓ_2 Norm. Cameron Musco, Christopher Musco, David P. Woodruff. *Preprint* 2021.

Error Bounds for Lanczos-based Matrix Function Approximation. Tyler Chen, Anne Greenbaum, Cameron Musco, Christopher Musco. *Preprint* 2021.

Linear and Sublinear Time Spectral Density Estimation. Vladimir Braverman, Aditya Krishnan, Christopher Musco. *Preprint* 2021.

Dynamic Trace Estimation. Prathamesh Dharangutte, Christopher Musco. *Preprint* 2021.

Correlation Sketches for Approximate Join-Correlation Queries*. Aécio Santos, Aline Bessa, Fernando Chirigati, Christopher Musco, Juliana Freire. *ACM Special Interest Group on Management of Data Conference (SIGMOD)* 2021.

Public Transport Planning: When Transit Network Connectivity Meets Commuting Demand*. Sheng Wang, Yuan Sun, Christopher Musco, Zhifeng Bao. *ACM Special Interest Group on Management of Data Conference (SIGMOD)* 2021.

Simple Heuristics Yield Provable Algorithms for Masked Low Rank Approximation. Cameron Musco, Christopher Musco, David Woodruff. *Innovations in Theoretical Computer Science (ITCS)* 2021.

Hutch++: Optimal Stochastic Trace Estimation. Raphael Meyer, Cameron Musco, Christopher Musco, David P. Woodruff. *SIAM Symposium on Simplicity in Algorithms (SOSA)* 2021.

Graph Learning for Inverse Landscape Genetics. Prathamesh Dharangutte, Christopher Musco. *AAAI Conference on Artificial Intelligence (AAAI) 2021*. Short version in NeurIPS 2020 AI for Earth Sciences workshop.

Fourier Sparse Leverage Scores and Approximate Kernel Learning. Tamás Erdélyi, Cameron Musco, Christopher Musco. *Conference on Neural Information Processing Systems (NeurIPS)* 2020. Invited for spotlight presentation (4% of submissions).

Near Optimal Linear Algebra in the Online and Sliding Window Models. Vladimir Braverman, Petros Drineas, Cameron Musco, Christopher Musco, Jalaj Upadhyay, David P. Woodruff, Samson Zhou. *IEEE Symposium on Foundations of Computer Science (FOCS)* 2020.

The Statistical Cost of Robust Kernel Hyperparameter Turning. Raphael A. Meyer, Christopher Musco. *Conference on Neural Information Processing Systems (NeurIPS)* 2020.

Low-Rank Toeplitz Matrix Estimation via Random Ultra-Sparse Rulers. Hannah Lawrence, Jerry Li, Cameron Musco, Christopher Musco. *International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*. 2020.

Sample Efficient Toeplitz Covariance Estimation. Yonina Eldar, Jerry Li, Cameron Musco, Christopher Musco. *ACM-SIAM Symposium on Discrete Algorithms (SODA)* 2020.

Fast and Space Efficient Spectral Sparsification in Dynamic Streams. Michael Kapralov, Aida Mousavifar, Cameron Musco, Christopher Musco, Navid Nouri, Aaron Sidford, Aaron Sidford. *ACM-SIAM Symposium on Discrete Algorithms (SODA)* 2020.

Understanding Filter Bubbles and Polarization in Social Networks. Uthsav Chitra, Christopher Musco. *ACM International Conference on Web Search and Data Mining (WSDM) 2020*. Preliminary version in KDD Workshop on Issues of Sentiment Discovery and Opinion Mining.

A Universal Sampling Method for Reconstructing Signals with Simple Fourier Transforms. Haim Avron, Michael Kapralov, Cameron Musco, Christopher Musco, Ameya Velingker, Amir Zandieh. *ACM Symposium on Theory of Computing (STOC)* 2019.

Learning Networks from Random Walk-Based Node Similarities. Jeremy Hoskins, Cameron Musco, Christopher Musco, Charalampos Tsourakakis. *Conference on Neural Information Processing Systems (NeurIPS)* 2018.

Eigenvector Computation and Community Detection in Asynchronous Gossip Models. Frederik Mallmann-Trenn, Cameron Musco, Christopher Musco. *International Colloquium on Automata, Languages, and Programming (ICALP)* 2018.

Minimizing Polarization and Disagreement in Social Networks. Cameron Musco, Christopher Musco, Charalampos Tsourakakis. *The Web Conference (WWW)* 2018.

Stability of the Lanczos Method for Matrix Function Approximation. Cameron Musco, Christopher Musco, Aaron Sidford. *ACM-SIAM Symposium on Discrete Algorithms (SODA)* 2018.

Recursive Sampling for the Nyström Method. Cameron Musco, Christopher Musco. *Conference on Neural Information Processing Systems (NeurIPS)* 2017.

Random Fourier Features for Kernel Ridge Regression: Approximation Bounds and Statistical Guarantees. Haim Avron, Michael Kapralov, Cameron Musco, Christopher Musco, Ameya Velingker, Amir Zandieh. *International Conference on Machine Learning (ICML)* 2017.

Input Sparsity Time Low-Rank Approximation via Ridge Leverage Score Sampling. Michael B. Cohen, Cameron Musco, Christopher Musco. *ACM-SIAM Symposium on Discrete Algorithms (SODA)* 2017.

Determining Tournament Payout Structures for Daily Fantasy Sports. Christopher Musco, Maxim Sviridenko, Justin Thaler. *SIAM Algorithm Engineering and Experiments (ALENEX)* 2017.

Principal Component Projection Without Principal Component Analysis. Roy Frostig, Cameron Musco, Christopher Musco, Aaron Sidford. *International Conference on Machine Learning (ICML)* 2016.

Randomized Block Krylov Methods for Stronger and Faster Approximate Singular Value Decomposition. Cameron Musco, Christopher Musco. *Conference on Neural Information Processing Systems (NeurIPS)* 2015. Invited for full oral presentation (1 of 15, < 1% of submissions). Also presented at the 2016 Copper Mountain Conference on Iterative Methods

Dimensionality Reduction for K-Means Clustering and Low Rank Approximation. Michael B. Cohen, Samuel Elder, Cameron Musco, Christopher Musco, Madalina Persu. *ACM Symposium on Theory of Computing (STOC)* 2015.

Principled Sampling for Anomaly Detection. Brendan Juba, Fan Long, Christopher Musco, Stelios Sidiropoulos, Martin Rinard. *Network and Distributed System Security Symposium (NDSS)* 2015.

Uniform Sampling for Matrix Approximation. Michael B. Cohen, Yin Tat Lee, Cameron Musco, Christopher Musco, Richard Peng, Aaron Sidford. *Innovations in Theoretical Computer Science (ITCS)* 2015.

Single Pass Spectral Sparsification in Dynamic Streams. Michael Kapralov, Yin Tat Lee, Cameron Musco, Christopher Musco, Aaron Sidford. *IEEE Symposium on Foundations of Computer Science (FOCS)* 2014, Special Issue SIAM Journal on Computing 2017.

Funding

NSF CAREER Award

Frontiers in Matrix Sketching.

2021-2026

PI, \$562,224

NSF III Medium

Dataset Search and Ranking for Data Augmentation and Explanation.

2021-2024

Co-PI (w/ Juliana Freire, NYU), \$1,093,195

DOE Office of Science Grant

Leverage Score Sampling for Parametric PDEs.

2021-2023

PI (w/ Chinmay Hegde, NYU), \$300,000

Current Student and Postdoc Advising

Aline Bessa. Postdoc, New York University.

2021 - present

Aarshvi Gajjar. Ph.D., New York University (with Prof. Chinmay Hedge).

2021 - present

Apoorv Singh. Ph.D., New York University.	2020 - present
Raphael Meyer. Ph.D., New York University.	2019 - present
Indu Ramesh. M.S., New York University.	2020 - present
Xinyu Luo. M.S., New York University.	2020 - present

Past Student Advising

Prathamesh Dharangutte. M.S., New York University.	2019 - 2021
John Shin. M.S., New York University.	2019 - 2020
Qilei Cai. Undergraduate, New York University.	2020
Ly Cao. Undergraduate, New York University.	2020
Charles Chan. Undergraduate, New York University.	2020
Denys Fenchenko. Undergraduate, New York University.	2020
Baizhou Hou. Undergraduate, New York University.	2020
Jason Soong. Undergraduate, New York University.	2020
Tristan Pollner. Undergraduate, Princeton.	2019

Ph.D. Thesis Committees

Francis Williams	Defended 2021.
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Teaching

Algorithmic Machine Learning and Data Science (NYU CS-GY 9223I) <i>Course Instructor</i>	Fall 2019, 2020, 2021
Introduction to Machine Learning (NYU CS-UY 4563) <i>Course Instructor</i>	Spring 2020
Advanced Algorithm Design (Princeton COS 521) <i>Course Instructor</i>	Fall 2018
Advanced Algorithms (MIT 6.854/18.415) <i>Teaching Assistant</i>	Spring 2016
Technical Communication Skills for Graduate Students (MIT 6.S977) <i>Workshop Leader</i>	Spring 2016

Mentorship and Outreach

SIAM/MathWorks Math Modeling Challenge <i>Problem Development Committee, Director of Technical Computing Judging</i>	2018 – present
Support SIAM's annual math modeling competition, which gives high school students an opportunity to apply mathematics, statistics, and data science to a real-world problem. Serve on committee in charge of writing practice and contest problems. Lead judging and award presentation for student Technical Computing prizes.	
MIT Graduate Communication Lab <i>Communication Adviser</i>	2015 – 2018
Helped to establish an EECS Communication Lab to provide over 700 graduate students with support in academic writing, oral presentation, job applications, and generally, in communicating effectively. Offered feedback, guidance, and coaching to individual students during weekly office hours.	

CSAIL Algorithms Office Hours

Member

2016 – 2018

Advised researchers, generally in applied fields, on framing and solving algorithmic problems arising in their work.

Research Science Institute at MIT

Summer Research Adviser

2014

Primary supervisor for a high school student participating in the prestigious RSI summer research program. Advisee's project, "Relaxation of a Concurrent Disjoint-Set", was selected as one of 10 program finalists.

Talks and Presentations

Mathematical Modeling and Coding in the Classroom

Mu Alpha Theta National Convention

July 2021

Randomized Algorithms for Probing the Spectrum of a Matrix

George Mason University, AI Tea Series

Apr. 2021

Sublinear Time Spectral Density Estimation

Workshop on Algorithms for Large Data

Aug. 2021

Cornell University, Scientific Computing and Numerics Seminar

Mar. 2021

Importance Sampling for Approximating High-Dimensional Kernel Matrices

Emory University, Numerical Analysis and Scientific Computing Seminar

Feb. 2021

Optimal Stochastic Trace Estimation

Stanford University, Operations Research Seminar

Oct. 2020

Structured Covariance Estimation

Yahoo! Research, Seminar

April 2020

Johns Hopkins University, Theory Seminar

Oct. 2019

Symposium on Discrete Algorithms (SODA)

Jan. 2020

From Randomized Linear Algebra to Randomized Functional Analysis

SIAM Mathematics of Data Science Conference

May 2020

University of Utah, Data Science Seminar

Jan. 2020

Rutgers University, DIMACS RandNLA Workshop

Sept. 2019

Learning Signals with Simple Fourier transforms

Symposium on Theory of Computing (STOC)

June 2019

Columbia University, Theory Seminar

Oct. 2018

UC Berkeley, Simons Institute Workshop

Sept. 2018

Princeton University, Theory Seminar

Sept. 2018

The Lanczos Method in Data Science: Challenges and Opportunities

SIAM Conference on Parallel Processing for Scientific Computing, Minisymposium

Feb. 2020

Georgetown University, CS Seminar

Feb. 2018

Symposium on Discrete Algorithms (SODA)

Jan. 2018

Unifying Random Features and Leverage Scores for Kernel Matrix Approximation

Institute for Advanced Study, Computer Science and Discrete Mathematics Seminar

Feb. 2018

Harvard University, guest lecture in "Sketching Algorithms for Big Data"

Nov. 2017

Microsoft Research, Redmond

Sept. 2017

Sketching and Sampling Methods for Large Scale Linear Algebra

Twitter, Cambridge	<i>July 2017</i>
Recursive Sampling for the Nyström Method	
Neural Information Processing Systems (NeurIPS) poster session	<i>Dec. 2017</i>
New England Machine Learning Day poster session	<i>May 2017</i>
Algorithms for Determining Tournament Payout Structures	
Meeting on Algorithm Engineering and Experiments (ALENEX)	<i>Jan. 2017</i>
Introduction to Linear Sketching	
MIT, Theory Retreat	<i>Sept. 2016</i>
Principal Component Projection without Principal Component Analysis	
International Conference on Machine Learning (ICML)	<i>June 2016</i>
Ridge Leverage Score Sampling	
National Institute of Informatics, Shonan Meeting	<i>July 2016</i>
MIT, Algorithms & Complexity Seminar	<i>July 2016</i>
University of Utah, Data Group Meeting	<i>Jan. 2016</i>
Randomized Block Krylov Methods	
Neural Information Processing Systems (NeurIPS) poster session	<i>Dec. 2015</i>
Large Scale Column Subset Selection	
Yahoo Labs, Science Week poster session	<i>July 2015</i>
Dimensionality Reduction for k-Means Clustering	
IBM T.J. Watson, Mathematical Sciences research seminar	<i>Aug. 2015</i>
Symposium on Theory of Computing (STOC)	<i>June 2015</i>
MIT, Theory Lunch	<i>June 2014</i>
Principled Sampling for Anomaly Detection	
MIT, Defense Advanced Research Projects Agency site visit	<i>Mar. 2015</i>
Network and Distributed System Security Symposium (NDSS)	<i>Feb. 2015</i>
Uniform Sampling for Matrix Approximation	
Yahoo Labs, Scalable Machine Learning research seminar	<i>June 2015</i>
Single Pass Spectral Sparsification in Dynamic Streams	
MIT, Annual Sublinear Algorithms Day poster session	<i>Apr. 2015</i>
Harvard University, Theory Seminar	<i>Nov. 2014</i>
Foundations of Computer Science (FOCS)	<i>Oct. 2014</i>

Service

Organization.....	
SIAM Annual Meeting Minisymposium Organizer	<i>2021</i>
SIAM Mathematics of Data Science Minisymposium Organizer	<i>2020</i>
NYU Tandon Algorithmic Data Science Reading Group	<i>2019 - present</i>
Department Service.....	
Faculty Search Committee	<i>2019,2021</i>
Online Faculty Oversight Committee	<i>2021 - present</i>
Ph.D. Admissions Committee	<i>2019 - present</i>

Conference Programming.....
 NeurIPS Area Chair 2019,2021
 Conference Reviewing.....
 STOC, FOCS, SODA, ICALP, ITCS, ICML, NeurIPS, AAI, SOCG, ESA, APPROX, RANDOM, STACS, IWOCA, WALCOM, NDSS
 Journal Reviewing.....
 Journal of the ACM, SIAM Journal on Computing, ACM Transactions on Algorithms, Theoretical Computer Science, Journal of Machine Learning Research, SIAM Journal on Matrix Analysis and Applications, SIAM Journal on Scientific Computing, SIAM Journal on Mathematics of Data Science, Science Advances, Mathematical Programming, IEEE Transactions on Signal Processing, IEEE Transactions on Knowledge and Data Engineering, International Journal of Foundations of Computer Science, IEEE Journal on Selected Areas in Information Theory, BIT Numerical Mathematics
 Grants Panels.....
 NSF Division of Information and Intelligent Systems (2019)
 Notables.....
 ICML 2018 Outstanding Reviewer, ICML Expert Review 2021

Invited Workshops and Events

Rutgers, DIMACS
Invited participant 2019
 Workshop on Randomized Numerical Linear Algebra, Statistics, and Optimization.
UC Berkeley, Simons Institute for the Theory of Computing
Invited participant 2018
 Workshop on Randomized Numerical Linear Algebra and Applications.
Japan National Institute of Informatics
Invited participant 2016
 Shonan meeting on Recent Advances in Randomized Numerical Linear Algebra.

Honors and Awards

National Science Foundation 2014-2017
Graduate Research Fellowship, recipient.
Yale University 2012
Cum Laude, Distinction in Both Majors.

Other Professional Experience

Redfin **Seattle, WA**
Software Engineer, Data Team 2012 – 2014
Elysium Digital **Cambridge, MA**
Summer Consultant 2011
Amicus **New Haven, CT**
Software Developer 2010 – 2011